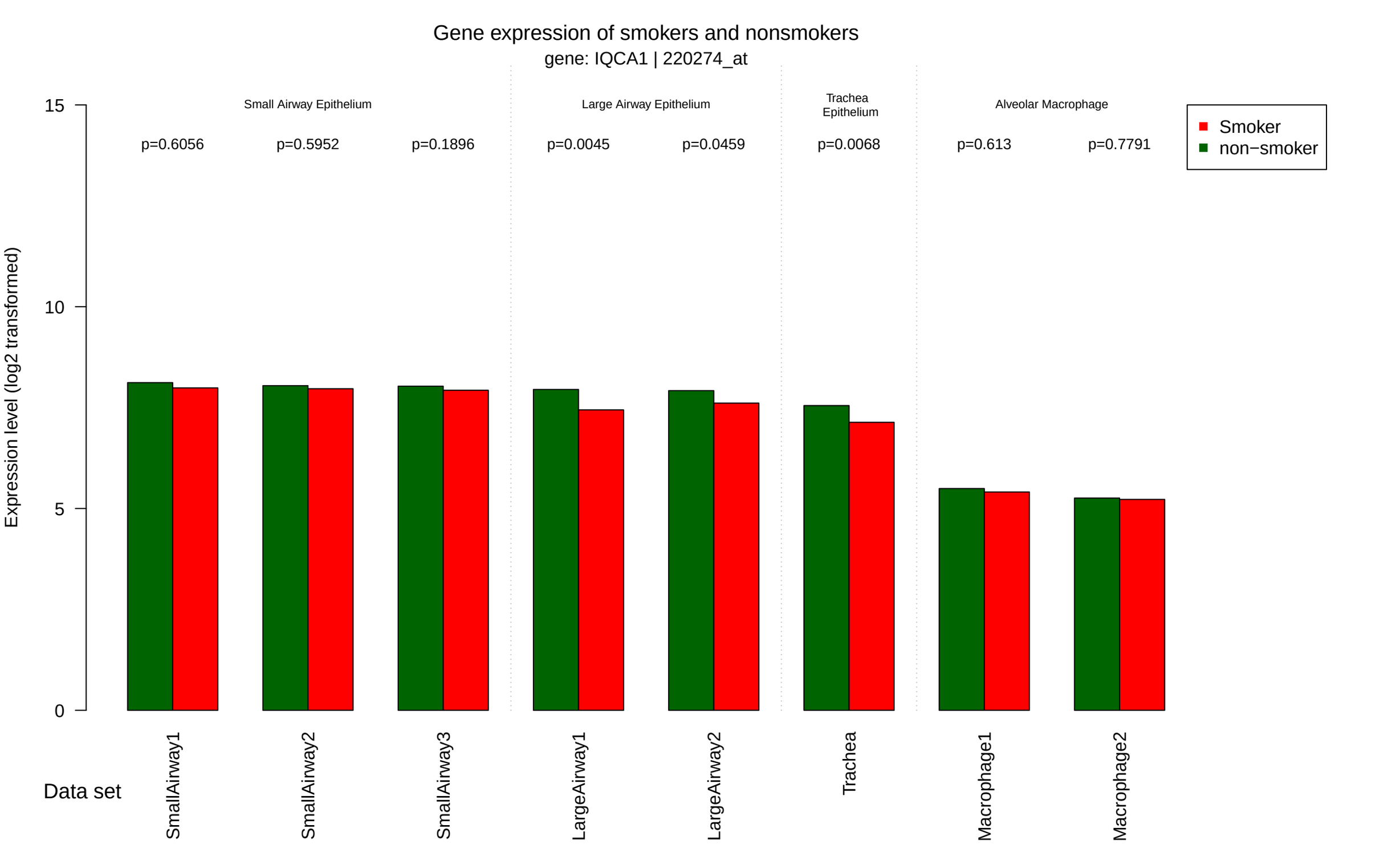
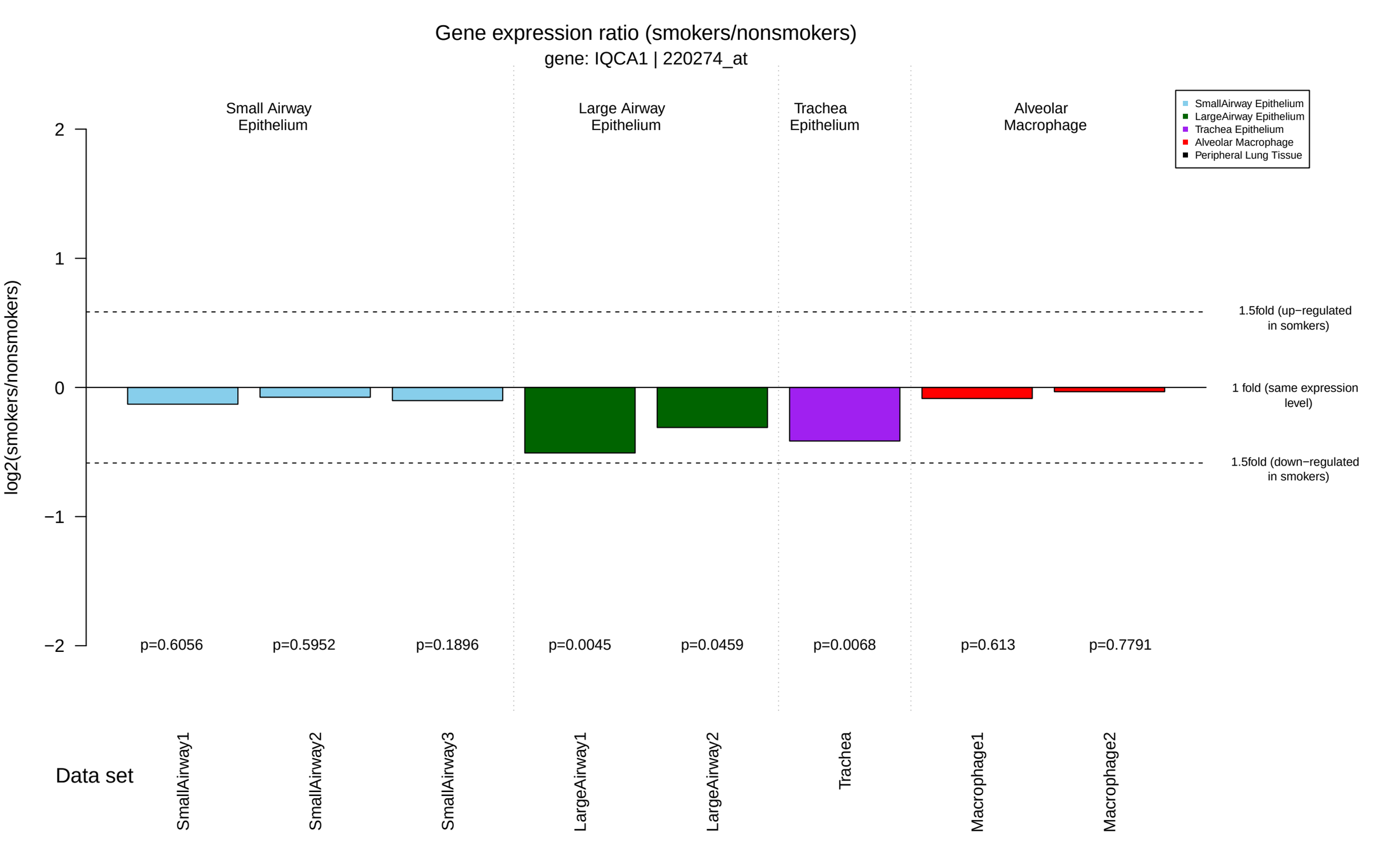
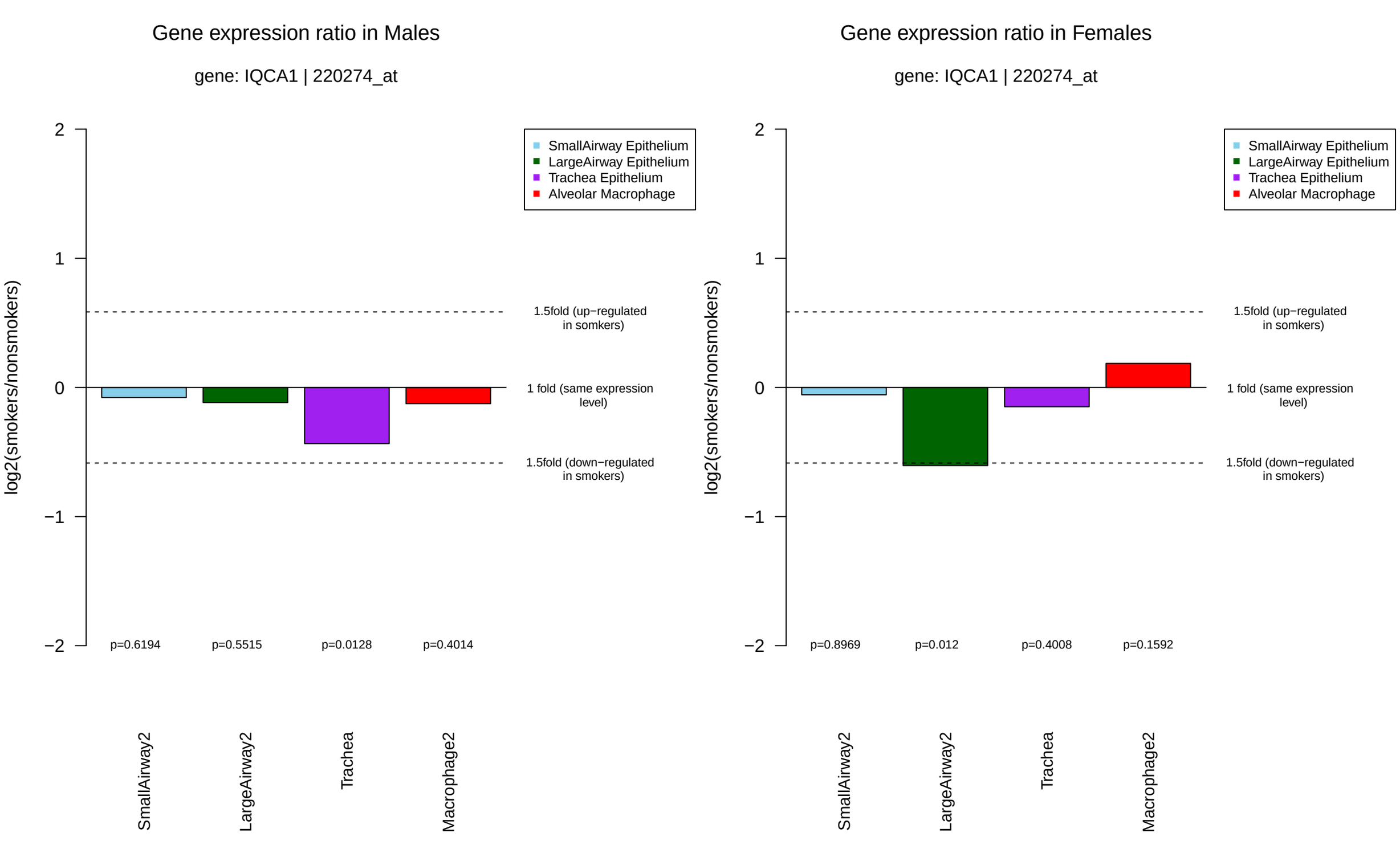
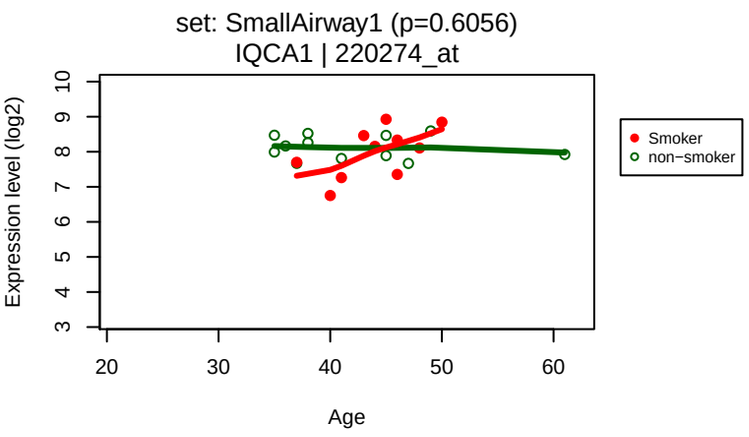
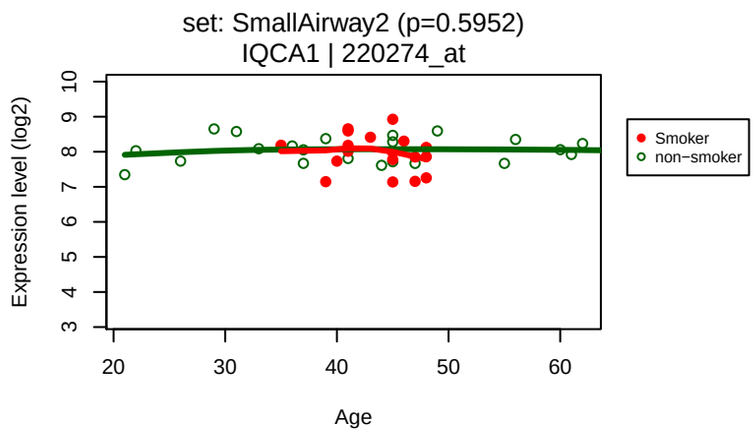
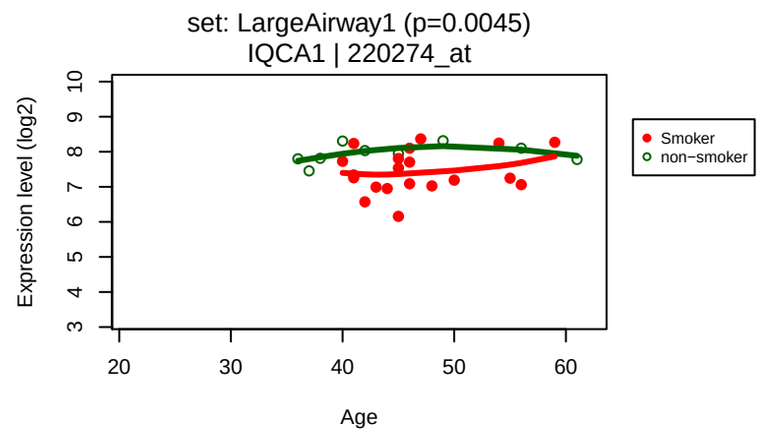

Gene expression of smokers and nonsmokers
gene: IQCA1 | 220274_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6056
p=0.5952
p=0.1896
p=0.0045
p=0.0459
p=0.0068
p=0.613
p=0.7791
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

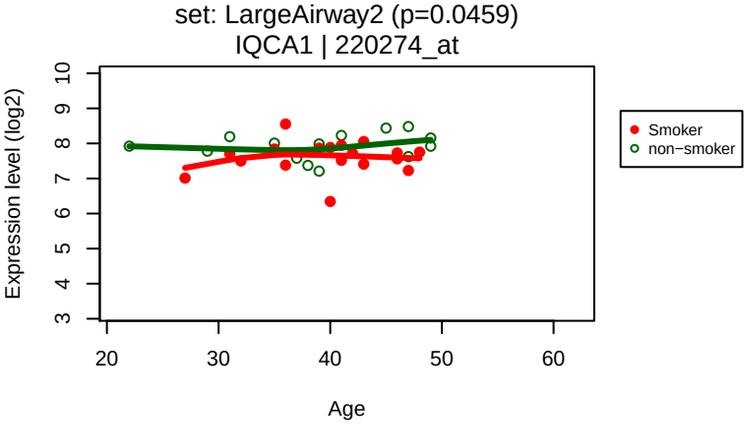

Gene expression ratio (smokers/nonsmokers)
gene: IQCA1 | 220274_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6056
p=0.5952
p=0.1896
p=0.0045
p=0.0459
p=0.0068
p=0.613
p=0.7791
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

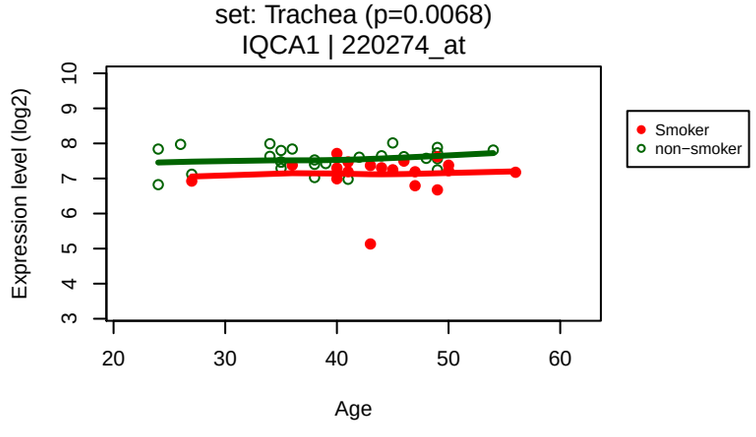

Gene expression ratio in Males
gene: IQCA1 | 220274_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6194
p=0.5515
p=0.0128
p=0.4014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IQCA1 | 220274_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8969
p=0.012
p=0.4008
p=0.1592
SmallAirway2
LargeAirway2
Trachea
Macrophage2

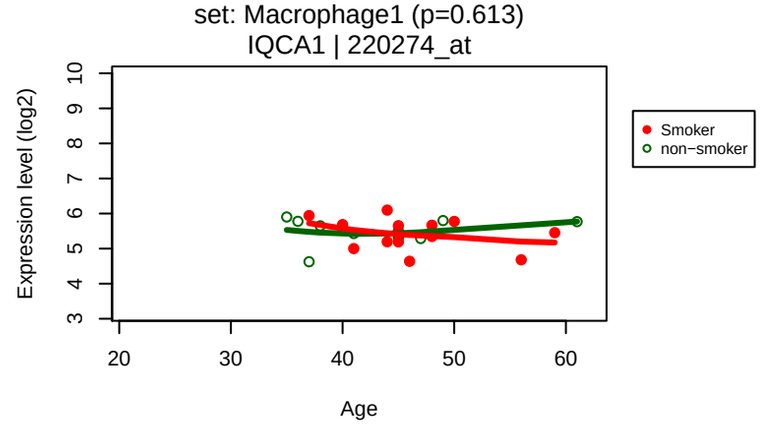

set: SmallAirway1 (p=0.6056)
IQCA1 | 220274_at
Expression level (log2)
Age
Smoker
non−smoker

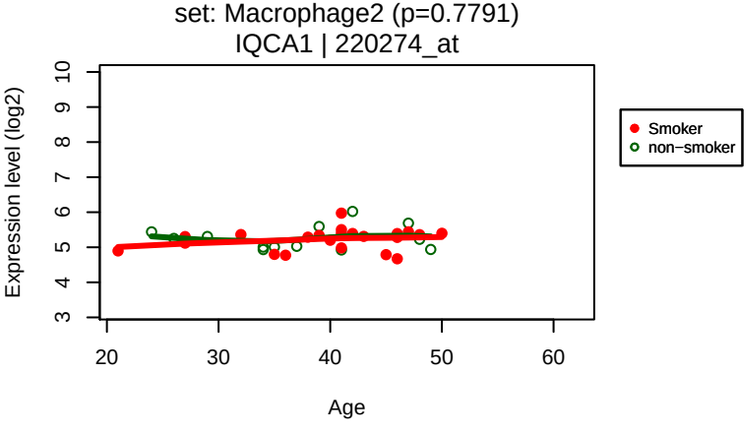

set: SmallAirway2 (p=0.5952)
IQCA1 | 220274_at
Expression level (log2)
Age
Smoker
non−smoker

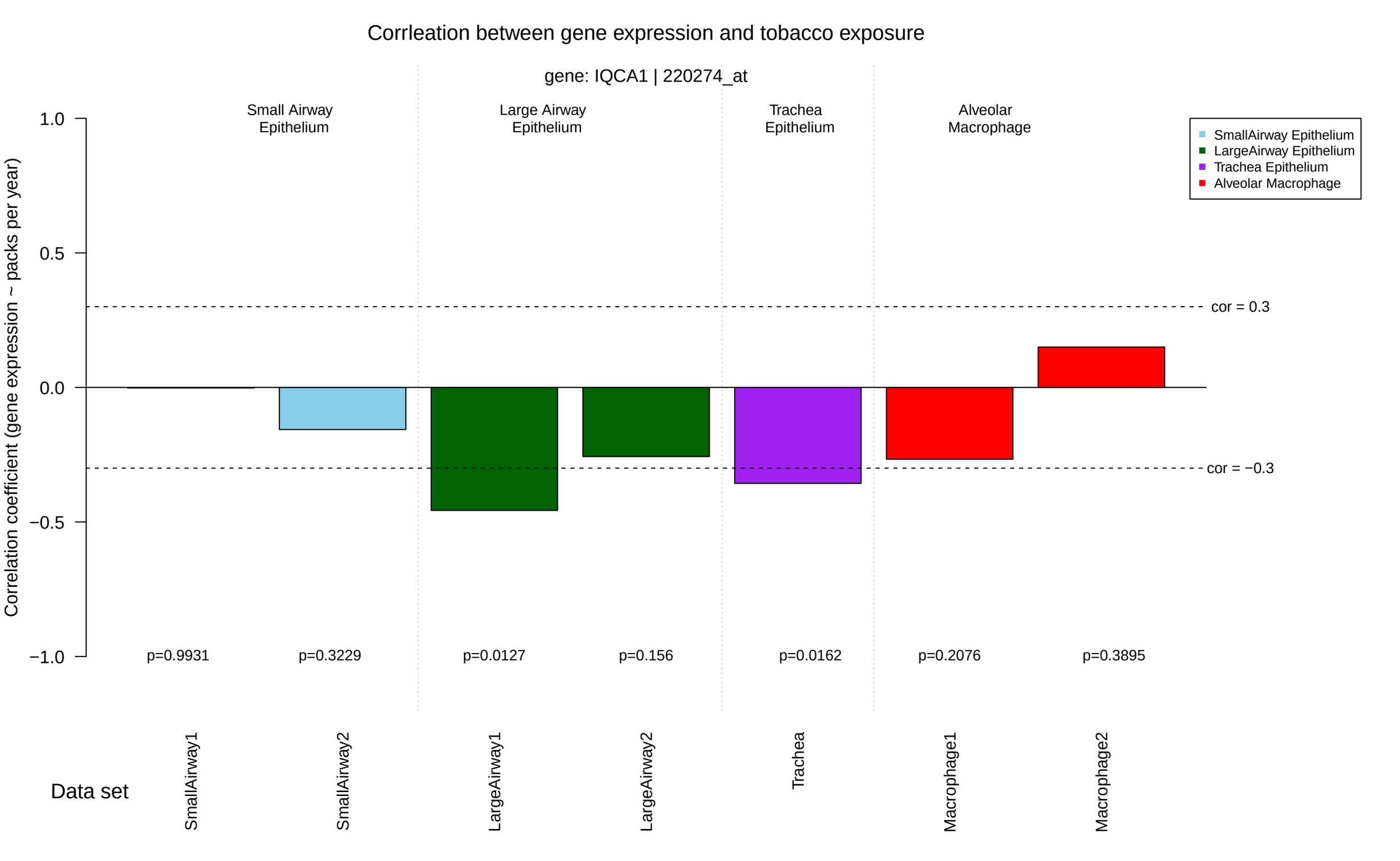

set: LargeAirway1 (p=0.0045)
IQCA1 | 220274_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0459)
IQCA1 | 220274_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0068)
IQCA1 | 220274_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.613)
IQCA1 | 220274_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7791)
IQCA1 | 220274_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IQCA1 | 220274_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9931
p=0.3229
p=0.0127
p=0.156
p=0.0162
p=0.2076
p=0.3895
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2